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A hybrid approach to extraction of knowledge from scientific texts based on large language models and domain dictionaries

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Abstract—The vast information landscape of the Internet constitute a significant challenge for extracting valuable content. The lack of standardized data models and structures necessitates ad hoc solutions, often requiring expert knowledge that developers may lack. While Large Language Models (LLMs) hold promise for addressing this challenge, their susceptibility to AI hallucinations and inaccuracies necessitates ongoing research.

This paper introduces an expert-centered approach to web-based knowledge extraction. We propose a hybrid method that combines domain-specific dictionaries with LLMs, exemplified by GigaChat, to improve the accuracy of method group definition. Our system incorporates explainability, allowing users to understand the reasoning behind method assignments. Furthermore, user-driven training is facilitated by allowing users to select theories and highlight relevant keywords, enhancing the system's learning capabilities.

Evaluations demonstrate the effectiveness of this approach, achieving an impressive F1-score of 89%. This research contributes to the ongoing effort to develop robust and accurate knowledge extraction systems for navigating the ever-expanding landscape of online information.

Index Terms—Large Language Model, Ontology, Fact extraction, Acoustic

I. INTRODUCTION

A common problem characteristic of modeling, simulation and analysis of complex physical processes is insufficient theoretical bases to cover large sets of various conditions parameterizing the processes. This includes cases when domains of different models overlap with specific particularities or, on the contrary, the parameter sets are such that neither of the existing models are well-defined. For instance, it can be problematic to make an a priori choice of a mathematical model and respective approximations to analyze a physical field at boundaries between near and far field zones, to choose

the correct precisions for real-valued constants and accuracy requirements to operations, adequate hardware and software, et cetera.

As a result, practical implementation of mathematical models and modeling techniques is intrinsically coupled with ad hoc expert decision making, often constrained by development time, and often this can yield inaccurate results and a need to iteratively modify the whole system until the desired performance of the model is achieved.

This paper focuses on such a class of problems which take place in architectural acoustics. Despite the widespread use of sound modeling methods, their classification with respect to large sets of parameters remains poorly defined in scientific and technical literature. Existing recommendations [9]–[11] offer solutions only for narrow sets of specific problems and conditions, often with omissions of specific cases such as those conditioned upon boundaries of applicability of several mathematical models, known as wave, geometrical and statistical acoustics. This incompleteness leads to a known lack of comprehensive understanding about the best-fit mathematical model, conditioned upon context, to use in order to obtain results with required accuracy and limits on use of computational resources, which is a current problem in many subject areas [12]–[14].

A way to mitigate this issue could be a use of automated expert system employed to facilitate decision making regarding the correct, i.e. the most efficient, way to model and simulate the studied physical process. To address this challenge, a knowledge bank is developed to provide a clear classification of modeling methods. However, the constant evolution of these methods necessitates a dynamic and robust system for knowledge acquisition.

This article proposes a novel approach to automate the extraction of data on acoustic methods from scientific articles, written in English, utilizing two methods: domain dictionaries and large language model, enabling the efficient identification and categorization of acoustic modeling techniques.

II. RELATED WORKS

Despite a large number of studies, there is no consensus on which method of knowledge extraction should be used.

Traditional approaches to information extraction from scientific texts have relied heavily on dictionary-based methods. Systems like Autonet utilize keyword extraction to identify potential phrases, but they often struggle with texts that employ non-standard ways of conveying information, leading to errors [1]. While the introduction of neural networks has mitigated some of these issues, dictionary-based methods still suffer from interference and limited coverage of preexisting dictionaries.

The recent emergence of large language models (LLMs) has opened new avenues for information extraction. Our proposed method leverages the capabilities of LLMs like GPT-3 and Llama-2, demonstrating high performance of information extraction from materials on both sentence-level and document-levels [2]. This method allows users to train bespoke models without requiring extensive knowledge of LLM internals, simplifying the process for researchers with limited knowledge of natural language processing (NLP).

While LLM-based methods offer significant advantages, they also have certain limitations. One challenge lies in ensuring valid output schema formatting during the generation step. LLMs may produce unparseable sequences, particularly when the inference token limit is below 512 tokens and the schema is JSON or XML, which typically requires a larger number of tokens for accurate parsing. This issue can be mitigated by increasing the token limit and, as LLMs continue to evolve, the token limitation is expected to become less of a concern.

Another limitation is the tendency of LLMs to hallucinate [3]–[6], that is to generate information not present in the input text. Such situations can arise for a variety of reasons, the most common being the lack of up-to-date knowledge of the domain [8], making the creation of specialized corpora and dictionaries relevant.

III. METHODOLOGY

A. Common workflow

Common workflow is presented as Algorithm 1 and defined as the following steps. Before extracting knowledge about the method, it is necessary to process the input data. The PDF file is converted to raw text format. The text is evaluated in parallel by two systems using dictionaries and the language model.

The user is then provided with the result of the evaluation and an explanation of why this decision was made by the system. However, the final decision is made by the user as there is a possibility of erroneous conclusions of both systems. If such a situation occurs, the user will be asked to show on

Algorithm 1: The general algorithm of the system.

Input: Article file_pdf in English in PDF format.
Domain specific dictionary input_dictionary.
Output: A report total_theory with the relevant theory.

```

1 file_txt ← convert_pdf_to_txt (file_pdf)
  /* Dictionary-based analysis */
2 dictionary_theory ←
  dictionary_analysis (input_dictionary,
    file_txt) /* LLM-based analysis */
3 LLM_theory ← LLM_analysis (file_txt)
4 LLM_expl ← get_explanation (LLM_theory)
  /* User picks acceptable theory,
    either dictionary_theory or LLM_theory,
    based upon statistics w.r.t.
    dictionary_theory and the LLM theory
    LLM_theory with reasoning in
    LLM_expl, or neither, if none is
    acceptable: */
5 user_choice ← get_user_choice_
  upon (statistics (dictionary_theory),
    LLM_theory, LLM_expl)
6 if user_choice is not dictionary_theory or
  user_choice is not LLM_theory then
7   total_theory ← user_theory
8   user_criteria ← get_user_criteria ()
  /* obtain the keywords by which
    the user identified the theory */
9   extend (input_dictionary, user_criteria)
10  send_learning_prompt (user_theory,
    user_criteria)
11 end
12 return total_theory

```

the basis of which words or phrases the conclusion that the method belongs to a certain theory was made. After that, the domain dictionary will be updated with new words, and a prompt with the specified theory selection criteria will be sent to the LLM-based subsystem, which will allow to refine the language model and make further answers more accurate.

The details upon the approaches to text analysis which are employed in the algorithm 1 are described in the subsequent sections of the paper.

B. Dictionary-based algorithm

The dictionary-based system implements the algorithms 2 and 3. The processed text undergoes further refinement, including the removal of non-printable characters, segmentation into sentences and individual words, and stemming to identify word roots. These steps are crucial for reducing text size and enhancing word search efficiency.

Text reduction occurs in two phases. Initially, the references section and non-methodological subsections (e.g. funding in-

Algorithm 2: The general algorithm of the dictionary-based system.

Input: `file_txt`— an English article in raw text format; `confidence`— confidence in frequency difference.

Output: A report with the number of words of the relevant theory and probable theory.

```

1 dictionary_analysis(file_txt):
2   |   preprocessed_file ← Preprocess(file_txt)
3   |   main_text ←
4   |   |   Get_main_text(preprocessed_file)
5   |   |   filtered_text ←
6   |   |   |   filter_sentences(main_text)
7   |   |   doc_stat ←
8   |   |   |   find_theory_words(filtered_text,
9   |   |   |   |   input_dictionary, confidence)
10  |   |   report ← create_report(doc_stat)
11  |   |   return report
12 end

```

formation, acknowledgments) are eliminated. Subsequently, meaningful sentences are filtered using a common dictionary.

The critical stage of method classification involves a domain-specific terminology database, referred to as acoustic dictionary. This process takes the refined text containing key sentences as input and outputs an array of words categorized by acoustic theory associated with the words (cf. Algorithm 3).

Finally, a comprehensive report is generated, detailing the words identified in the text, their theoretical associations, and the total word count for each acoustic theory.

It was not possible to identify any unambiguous features for recognizing qualitative methods, except for the word "qualitative". Therefore, the method described by this word will be recognized as qualitative even if the values of other groups, that is a number of their uses in the source, are not zero.

In the process of analyzing the reports generated by the system based on dictionaries it was found that hybrid methods have relatively high word frequencies for two or more theories. In the proposed system, if the difference between two classes is not more than a user defined percentage (in the proposed system, the default value is 70%), then the method is considered as hybrid.

C. LLM-based algorithm

The system interacts with the GPT model through web requests using GigaChat API [7]. Prompts are generated in natural language using additional parameters presented on fig. 1.

temperature parameter: the higher the value, the more random the model response will be. If the temperature value is greater than two, the set of tokens in the model

Algorithm 3: The algorithm of classification of methods.

Input: `filtered_text` — raw text of the paper to analyze; `input_dictionary` — domain specific dictionary; `confidence` — confidence in frequency difference.

Output: Tuple $\text{Stat} = \langle S, T \rangle$, where S — count of the domain words for every theory, T — probable class of the method (statistical, geometrical, wave, qualitative, hybrid).

```

1 find_theory_words(filtered_text,
2   |   input_dictionary, confidence):
3   |   // load every theory and its words
4   |   words_by_theory ←
5   |   |   get_domain_words(input_dictionary)
6   |   |   // stemming words for every theory
7   |   |   stemmed_domain_words ←
8   |   |   |   stem(words_by_theory)
9   |   |   foreach  $t_w \in \text{stemmed\_domain\_words}$  do
10  |   |   |   count ← count_words(filtered_text,  $t_w$ )
11  |   |   |   Stat [ $t_w$ ] ← (count)
12  |   |   /* sort array by count of words to
13  |   |   |   find potential hybrids */
14  |   |   sorted_data ← reverse_sort(Stat, count)
15  |   |
16  |   |   if  $\frac{\text{sorted\_data}[1][1] \cdot 100}{\text{sorted\_data}[0][1]} \geq \text{confidence}$  then
17  |   |   |   Stat ['hybrid'] ← 1 // hybrid method
18  |   |   end
19  |   |   T ← max(S)
20  |   |   return Stat
21 end

```

```

"temperature": 1,
"top_p": 0.1,
"stream": False,
"max_tokens": 5000,
"repetition_penalty": 1

```

Fig. 1. Parameters for LLM

response may be excessively random. The temperature value must be greater than zero.

top parameter is used as an alternative to *temperature* and specifies the probability mass of tokens to be considered by the model. Thus, if you pass the value 0.1, the model will only consider tokens whose probability mass is in the top 10

stream parameter specifies that messages should be passed piecemeal in a stream.

max_tokens sets the maximum number of tokens that will be used to create responses.

repetition_penalty sets number of word repetitions. A value of 1.0 is a neutral value. With a value greater than 1, the model will try not to repeat words.

To extract a theory from the GigaChat response, an algorithm similar to 3 is used, except that the scores of each theory are not compared to identify hybrid methods.

EXPERIMENT

To validate the accuracy of extracting information on acoustic methodologies used in electronic scientific materials, several experiments were conducted. The test set for these experiments consisted of a dataset comprising 30 English-language scientific publications in the field of acoustics. Prior to the experiments, these materials were evaluated and categorized by experts according to specific acoustic theories. The dataset was divided into two different groups of publications: those containing information both about the proposed acoustic method, where the theory is explicitly stated, and those without explicitly stating it. To assess the effectiveness of the approaches, F1-score was used, having previously calculated true positive and false positive indicators. Afterwards, the answer of each system was evaluated. The answer is the triplet $\langle T, H, Q \rangle$, where T is the proposed theory, H is whether the method is hybrid, Q is whether the method is qualitative. Points are awarded for each correct answer from these three categories.

To assess the effectiveness of the approach, Precision indicators (1), recall (2), accuracy (3) and F1-score (4) were used, having previously calculated true positive and false positive indicators.

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (1)$$

where TP is True Positive, FP is False Positive.

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (2)$$

where FN is False Negative.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (3)$$

where TN is True Negative.

$$\text{F1-score} = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} \quad (4)$$

Cases in which the systems showed different results for different elements of the triplet $\langle T, H, Q \rangle$ were evaluated using formulas (5), (6) and (7) respectively.

$$\text{T-measure} = \frac{DT}{A}, \quad (5)$$

where DT is a number of different cases corresponding to the criterion "theory" and A is a number of articles.

$$\text{H-measure} = \frac{DH}{A}, \quad (6)$$

where DH is count of different cases for criterion "hybrid".

$$\text{Q-measure} = \frac{DQ}{A}, \quad (7)$$

where DQ is count of different cases for criterion "qualitative method".

These obtained values of accuracy are summarily shown in the table I.

TABLE I
THE RESULTS OF THE EXPERIMENT.

Criteria	Our approach
Precision	0.87
Recall	0.94
Accuracy	0.81
F1-score	0.89
T-measure	0.22
H-measure	0.13
Q-measure	0.26

The most common error for the LLM approach was theory selection, particularly for articles that did not explicitly state the method, and determining whether the method presented was qualitative. The most common error of the dictionary-based approach was identifying hybrid methods (falsely positive scores were noted).

CONCLUSION

The paper considers a system for automated extraction of knowledge from scientific texts based on the example of the subject area "Architectural Acoustics". To solve this task, a hybrid method based on the use of large language models (LLM) and domain dictionaries is used. The use of this approach is justified by the fact that LLM-based tools are not always effective in correctly classifying and extracting information, and there is a risk of AI hallucinations.

Our experiment has demonstrated that automated data extraction of acoustic methods from English-language scientific articles can significantly reduce the cost of populating a knowledge base with new rules. The combination of the two approaches described above showed an efficiency of 89%. This high efficiency is due to the fact that the systems compensate each other's deficiencies. However, it is evident that human involvement cannot be entirely eliminated from this process.

This convergence of results lends credibility to our approach and highlights the potential of automated data extraction. However, the criteria for identifying qualitative and hybrid methods need to be further refined in order to achieve greater accuracy, which is a subject of future work. The software implementation of the described system and set of papers are available in the repository at <https://gitlab.com/130120013/dictionarybasedknowledgeextractor>.

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